Samsung Innovation Campus

Enabling People Together



Tribhuvan University

Pulchowk Campus

SIC-AI Capstone Project

Mid-Term Progress Report

Speaker-Aware Noise Cancellation and Target Speech Enhancement

Submitted By

Austina Aryal (079BEI010)
Prashamsa Aryal (079BEI024)
Sostika Shrestha (079BEI041)

Submitted To

[TODO: insert the supervisor/mentor's full name and department]

November 2026 (Kartik 2083)

Abstract

We are building a target speaker extraction system for our SIC-AI capstone. The input is a single-channel recording that may contain the target speaker, another talker and background noise, together with a short enrollment recording of the target; the output should contain only the target's voice. The proposal combined a frozen ECAPA-TDNN speaker encoder with a causal Conv-TasNet/BSRNN separator. During implementation we dropped Conv-TasNet and kept a speaker-conditioned BSRNN with 16.83 M parameters. A frozen ECAPA-TDNN produces a 192-dimensional embedding, which is added once to the band features and injected through FiLM in each of the eight dual-path layers. The model is trained on on-the-fly mixtures of LibriSpeech speech and DEMAND noise with a combined SI-SDR and multi-resolution STFT loss.

Training finished all 100 epochs. Validation SI-SDRi on unseen speakers rose from 8.1 dB at epoch 33 to a best of 11.84 dB at epoch 97. On the 1200-mixture test set, the earlier-checkpoint evaluation gives a 10.8 dB SI-SDR gain and raises STOI from 0.77 to 0.87 on non-clean mixtures; a Whisper check on the final checkpoint lowers word error rate from 38.3% to 18.1%. The model chooses the wrong speaker in 7.7% of interferer mixtures, mostly when enrollment is short. It is non-causal, and the test tables still need to be rerun with the final checkpoint. Remaining work covers that rerun, the conditioning ablation, an unseen-noise test and CPU measurements.

Contents

1 Introduction **4**
1.1 Problem and motivation . . . 4
1.2 Objectives . . . 4
1.3 Scope of this report . . . 4

2 Progress and Changes from the Proposal **4**
2.1 Design changes . . . 4

3 Background and Related Work **6**

4 Current System Design **6**
4.1 Overview . . . 6
4.2 Front end and band split . . . 6
4.3 Speaker branch and conditioning . . . 7
4.4 Dual-path layers and mask decoder . . . 7

5 Data and Mixture Simulation **7**

6 Training Setup **8**
6.1 Loss . . . 8
6.2 Optimisation . . . 9

7 Results **9**
7.1 Training progress . . . 10
7.2 Test-set metrics . . . 10
7.3 Qualitative examples . . . 11
7.4 Speaker confusion . . . 11
7.5 Downstream intelligibility . . . 11
7.6 Efficiency . . . 12
7.7 Own-recording demonstration . . . 13

8 Challenges and Limitations **13**

9 Remaining Work and Revised Plan **14**

10 Conclusion **14**

A Reproducibility Notes **16**

B Additional Test Breakdowns **16**

1 Introduction

1.1 Problem and motivation

Conventional noise suppression removes environmental noise but treats every human voice as signal. In a call or a meeting, a second person talking nearby is not noise in that sense, yet is still unwanted. Target speaker extraction (TSE) handles this by conditioning the enhancement network on an enrollment recording of the desired speaker, so that only that voice is kept [1, 2]. Because the output is tied to one enrolled speaker, TSE avoids the permutation problem of blind speech separation and does not need to know how many speakers are present.

1.2 Objectives

The proposal set three objectives:

1. design and implement a target-speaker extraction system that combines a pretrained speaker encoder with a separator network;
2. evaluate quality, generalisation and efficiency (SI-SDR, STOI, PESQ, unseen speakers and noise, model size and compute);
3. deliver a reproducible training and evaluation framework.

1.3 Scope of this report

This report records what has been completed against the proposal timeline, how and why the design changed, the current system and data, the results so far, and the remaining work and risks.

2 Progress and Changes from the Proposal

Table 1 compares the work packages in the proposal timeline with their current status. Items in red are details that the supplied sources do not establish.

The main achievements so far are: a speaker-conditioned BSRNN with 16.83 M parameters, trained for the full 100 epochs, with a best validation SI-SDRi of 11.84 dB at epoch 97 on unseen speakers; an earlier-checkpoint test evaluation with a 10.8 dB SI-SDR gain, STOI rising from 0.77 to 0.87 and PESQ 2.20 on non-clean mixtures; a Whisper word error rate falling from 38.3% to 18.1%; a measured speaker-confusion rate of 7.7%; and a resumable training pipeline with cached embeddings, fixed manifests and an inference path that accepts a user's own mixture and enrollment recording.

2.1 Design changes

The largest change was the separator backbone. The proposal put Conv-TasNet first and BSRNN second. We later dropped the Conv-TasNet route, but the supplied sources contain no Conv-TasNet notebook or log: **[TODO: add the Conv-TasNet best SI-SDRi, number of epochs, configuration and a short reason it was dropped]**. BSRNN therefore became the model we trained to completion. Table 2 lists the other deviations.

Table 1: Status of the proposal work packages at mid-term.

Work package	Status	Notes
Literature review and dataset survey	**Done**	Reviewed in the proposal and extended with BSRNN and dual-path work (Section 3).
Audio pipeline and classical baseline	**Partial**	STFT/iSTFT pipeline done. **[TODO: state whether the classical baseline was implemented and give its result, or say that it was not run]**
Dataset preparation and mixture generation	**Done**	LibriSpeech and DEMAND manifests, fixed validation and test sets, on-the-fly scene sampler (Section 5).
ECAPA-TDNN encoder and conditioning ablation	**Partial**	Frozen ECAPA-TDNN with cached embeddings done; additive and FiLM conditioning implemented. The concat/FiLM/gating ablation is not done.
Separator backbones	**Partial**	Speaker-conditioned BSRNN trained for 100 epochs. Conv-TasNet was attempted and dropped; its details are missing (Section 2). TF-GridNet not attempted.
Curriculum training and augmentation	**Partial**	Randomised scenarios, SIR/SNR and levels. No curriculum schedule and no reverberation yet.
Evaluation and generalisation tests	**In progress**	Test metrics, scenario/SNR/SIR breakdowns, speaker confusion and WER are documented (Section 7); the final-checkpoint rerun, ablations and unseen-noise tests are pending.
Latency, CPU and memory optimisation	**Partial**	GPU cost measured offline (RTF 0.028, 16.83 M parameters). CPU speed and causal latency are not measured; the model is non-causal.
Final validation and documentation	**Pending**	This report is the first documentation milestone.

Table 2: Differences between the proposal and the current implementation.

Aspect	Proposal	Current	Reason
Separator	Causal Conv-TasNet; BSRNN as a stretch goal	Speaker-conditioned BSRNN, 8 dual-path layers	Conv-TasNet details are missing
Causality	Strictly causal (past and current frames)	Non-causal (bidirectional LSTMs)	Causal operation is deferred
STFT	512-point FFT, 160-sample hop (10 ms)	512-point FFT, 128-sample hop (8 ms)	Finer time resolution for BSRNN
Speaker encoder	ECAPA-TDNN, frozen versus fine-tuned	Frozen ECAPA-TDNN, cached embeddings	Saves GPU time and memory
Data	VoiceBank, DEMAND, MUSAN, DNS	LibriSpeech (1172 training speakers) and 16 DEMAND environments	Many clean speakers; one noise source
Reverberation	Limited RIR simulation	Not included yet	Listed as future work
Loss	SI-SDR + spectral + mask terms	SI-SDR (clamped at 30 dB) + 5× multi-resolution STFT	Waveform and spectral objective; mask term dropped
Test protocol	Speaker- and noise-disjoint sets	Speaker-disjoint; noise audio disjoint, environments shared	See Section 8

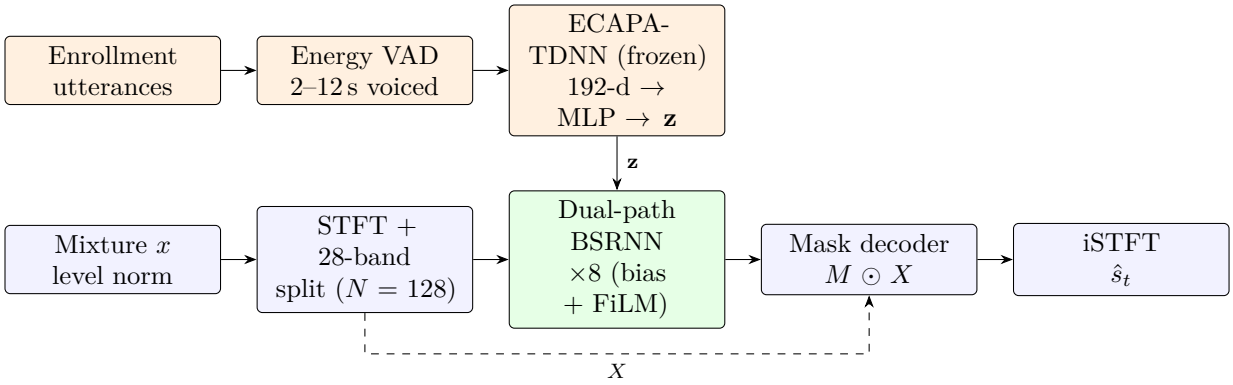


Figure 1: Speaker-conditioned BSRNN. The speaker vector $\mathbf{z}$ is added once to the band features and used by FiLM in every dual-path layer. The dashed path carries the mixture spectrogram to the mask multiplier.

3 Background and Related Work

Target speaker extraction. VoiceFilter [1] conditions a spectrogram mask estimator on a d-vector of the target speaker. SpeakerBeam-SS [2] uses lightweight Conv-TasNet variants with state-space modelling for low-latency operation. We use the same enrollment-conditioned formulation but a band-split time–frequency backbone.

Band-split RNN. BSRNN [3] splits the complex spectrogram into sub-bands, projects each band to a common feature size and alternates RNNs along time and along bands, in the spirit of dual-path models [4]. It was first used for music separation and later for efficient speech enhancement [5]. TF-GridNet [6] is a heavier alternative that we keep as a possible upper-bound reference.

Speaker embeddings and conditioning. ECAPA-TDNN [7] maps an utterance to a fixed-length speaker embedding; we use the SpeechBrain implementation [8]. Feature-wise linear modulation (FiLM) [9] scales and shifts feature channels from a conditioning vector and is our main speaker-injection mechanism.

Losses and metrics. We optimise scale-invariant SDR [10] with a multi-resolution STFT loss [11], and evaluate with SI-SDRi, STOI [12] and wideband PESQ [13]. The DNS Challenge [14] and DeepFilterNet [15] motivated the original efficiency goals.

4 Current System Design

4.1 Overview

Given a mixture $x = s_t + s_i + n$ (target, interferer, noise) and an enrollment embedding $\mathbf{e} \in \mathbb{R}^{192}$ of the target, the network predicts a complex mask M on the mixture spectrogram X and returns $\hat{s}_t = \mathrm{iSTFT}(M \odot X)$. Figure 1 shows the pipeline.

4.2 Front end and band split

Audio is processed at 16 kHz. The waveform is divided by its standard deviation, and the output is multiplied back, so the network sees level-independent input. An STFT with a 512-point Hann

window (32 ms) and a 128-sample hop (8 ms) gives 257 complex bins per frame; a 4 s training clip yields 501 frames. STFT, iSTFT and the loss are computed in fp32 even under mixed precision.

The 257 bins are grouped into 28 non-uniform bands: 8 bands of 125 Hz below 1 kHz, 12 bands of 250 Hz between 1 and 4 kHz, and 8 bands of 500 Hz between 4 and 8 kHz. The last band takes the one leftover bin. For each band, the real and imaginary parts are layer-normalised and linearly projected to $N = 128$ features, producing a tensor of shape $[B, 28, T, N]$. The fine low-frequency resolution follows the harmonic structure of speech; the coarser high-frequency bands cost less and carry less formant information.

4.3 Speaker branch and conditioning

Enrollment audio is cleaned with an energy-based voice activity detector (25 ms frames, 10 ms hop, threshold 30 dB below the 95th-percentile frame energy, floor −55 dB, 10-frame hang). The same VAD is used when building the cached enrollment banks and in `embed_enrollment` for a user's own recording. Voiced speech from several of the speaker's utterances is concatenated and cut to 2, 3, 5, 8 or 12 seconds (probabilities 0.15, 0.20, 0.30, 0.20, 0.15), then passed through a frozen SpeechBrain ECAPA-TDNN (`spkrec-ecapa-voxceleb`) [7, 8]. The 192-d embedding is L2-normalised and mapped by a small MLP ($192 \to 256 \to 256$, PReLU) to a speaker vector $\mathbf{z}$.

The vector enters the separator in two places. First, it is linearly projected to N dimensions and added once to the band features before the first layer. Second, every dual-path layer computes a per-channel scale and shift,

$$[\boldsymbol{\gamma}, \boldsymbol{\beta}] = W\mathbf{z} + \mathbf{b}, \qquad \mathbf{h}' = \mathbf{h} \odot (1 + \boldsymbol{\gamma}) + \boldsymbol{\beta}, \tag{1}$$

where $W \in \mathbb{R}^{2N \times 256}$. The FiLM weights and bias are zero-initialised, so training starts from an unconditioned BSRNN and learns to use the speaker cue gradually.

All enrollment embeddings are pre-computed into banks (10 recipes per training speaker, 6 per validation and test speaker, about 11.7 k training embeddings), so the encoder costs no GPU time or memory during training. A mixture's own target utterance is never part of its enrollment.

4.4 Dual-path layers and mask decoder

Each of the eight layers applies FiLM, then a time block (LayerNorm, BiLSTM along T for every band, projection, residual) and a band block (the same structure along the 28 bands for every frame). The time block gives long-range temporal context; the band block models cross-frequency structure at the low cost of a 28-step sequence.

The decoder is applied per band: LayerNorm, Linear$(N, 4N)$, Tanh, Linear$(4N, 4 \cdot \text{bins})$ and a GLU [16], which produces a real and an imaginary mask value for every bin. The mask is unbounded and complex, so it can change phase as well as magnitude: $\hat{S}(t, f) = M(t, f)X(t, f)$.

With $N = 128$, LSTM hidden size 256 and 8 layers, the model has **16.83 M parameters** and needs 23.9 GMACs for a 6 s clip (Section 7.6). This is much larger than the compact CPU model targeted in the proposal; reducing it is part of the remaining work (Section 9).

5 Data and Mixture Simulation

LibriSpeech [17] supplies clean read English at 16 kHz. We train on train-clean-100 and train-clean-360 (251 + 921 = 1172 speakers), validate on dev-clean (40 speakers) and test on test-clean

(40 speakers). Validation and test speakers never appear in training, so these sets measure generalisation to unseen voices. The validation set is 300 fixed mixtures (seed 1234) and the test set is 1200 fixed mixtures (seed 4321).

DEMAND [18] provides multichannel environmental recordings (kitchen, living room, park, street, metro, car, and so on). Sixteen of the 17 environments were available; `PCAFETER` did not download. Each recording is split by time: the first 80% for training, the next 10% for validation and the last 10% for testing. No noise audio is shared across splits, but the same 16 environments appear in all three.

Training data are generated on the fly. Every call samples a fresh scene with one of four scenarios: target plus interferer plus noise (probability 0.45), target plus interferer (0.25), target plus noise (0.25), or target alone (0.05). The target-to-interferer ratio is uniform in $[-5, 5]$ dB and the speech-to-noise ratio is uniform in $[-5, 15]$ dB. With probability 0.5 the interferer covers only 30–100% of the clip at a random offset. The mixture level is drawn from $[-35, -15]$ dBFS and passed through a peak limiter at 0.99. Training clips last 4 s and evaluation clips 6 s. One model therefore learns target extraction, plain denoising and leaving clean speech untouched. No training example removes the target entirely (Section 8).

6 Training Setup

6.1 Loss

For an estimate $\hat{s}$ and reference s, scale-invariant SDR [10] is

$$\text{SI-SDR}(\hat{s}, s) = 10 \log_{10} \frac{\|\alpha s\|^2}{\|\alpha s - \hat{s}\|^2}, \qquad \alpha = \frac{\langle \hat{s}, s \rangle}{\|s\|^2}, \tag{2}$$

computed after removing the mean of each signal. The training loss combines it with a multi-resolution STFT loss [11]:

$$\mathcal{L} = -\, w_1\, \mathbb{E}\big[\, \min(\text{SI-SDR},\, 30\,\text{dB})\big] + w_2\, \mathcal{L}_{\text{MR-STFT}}, \qquad w_1 = 1,\ w_2 = 5. \tag{3}$$

$\mathcal{L}_{\text{MR-STFT}}$ sums spectral-convergence and log-magnitude terms at three resolutions (window, hop) = (512,128), (1024,256) and (2048,512). The 30 dB clamp stops near-perfect examples, such as the clean scenario, from dominating the gradient. As a consistency check, a logged loss of about -5 with a training SI-SDR of about 11 dB implies an MR-STFT term of about 1.2.

6.2 Optimisation

Table 3: Training configuration of the main run.

Item	Value
Model	BSRNN, $N = 128$, hidden 256, 8 layers, 28 bands (16.83 M parameters)
Optimiser	AdamW [19], weight decay 0.01
Learning rate	2000-step linear warm-up to 5×10^{-4}, cosine decay [20] to 10^{-5} over 100k steps
Batch	8 clips × 4 s
Epoch definition	1000 steps (data generated on the fly); 100 epochs = 100k steps = 800k scenes
Gradient clipping	global norm 5.0
Precision	fp16 autocast with GradScaler [21]; STFT, iSTFT and loss in fp32
Memory saving	Gradient checkpointing [22] on every dual-path layer
Safety	Non-finite-loss batches are skipped and counted
Checkpointing	`_last.pt` every epoch (model, optimiser, scaler, log); `_best.pt` on best validation SI-SDRi
Outcome	100 epochs completed; best validation SI-SDRi 11.84 dB at epoch 97
Hardware	NVIDIA L4 (Google Colab)
Speed	0.99 s/step, peak GPU memory 3.2 GiB, about 17.4 min/epoch (about 29 h for 100 epochs)

Training ran on Colab, so the pipeline is built to resume: manifests and embedding banks live on Google Drive, and the log shows a successful resume from epoch 33. Figure 2 plots the run after that resume.

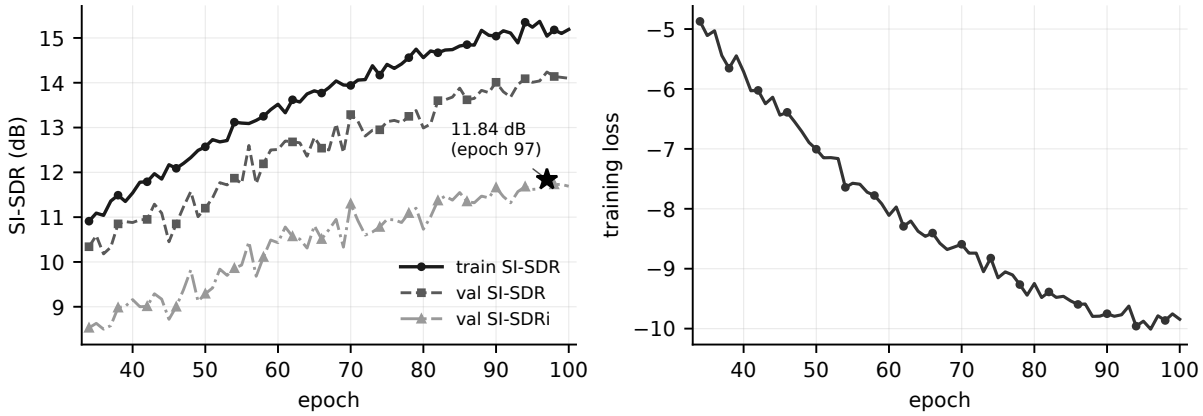


Figure 2: Training and validation SI-SDR, validation SI-SDRi (star: best epoch) and the combined loss for epochs 34–100.

7 Results

Training finished all 100 epochs. The best validation checkpoint (`tse_bsrnn_big_best.pt`) was reached at **epoch 97**, with a validation SI-SDRi of **11.84 dB** on unseen speakers. The fixed test set contains 1200 mixtures from 40 unseen speakers.

The test SI-SDR, STOI and PESQ numbers in this section come from a metrics file produced with an *earlier* best checkpoint, before training finished. They are historical results, not final-checkpoint test results. Speaker confusion, the WER check, the efficiency measurements and the six qualitative examples use the final checkpoint; on those six examples the final checkpoint is 0.1–1.4 dB better. **[TODO: rerun scripts/colab_rerun_final_eval.py with the epoch-97 checkpoint and replace Tables 4, 5 and 10, plus Appendix Table 11]**

7.1 Training progress

After the resume at epoch 33 (validation SI-SDRi then 8.12 dB), validation SI-SDRi rose from 8.5 dB at epoch 34 to 11.7 dB at epoch 100, with a best value of 11.84 dB at epoch 97. The training loss fell from −4.9 to −9.8 and training SI-SDR from 10.9 to 15.2 dB. Over the last ten epochs the train–validation SI-SDR gap averaged about 1.1 dB, so there is no strong overfitting. The curve flattens as the learning rate anneals towards 10^{-5}: from epoch 90 onward, validation SI-SDRi stayed between 11.3 and 11.8 dB. No batches were skipped for non-finite loss in the final epochs. Swings of a few tenths of a dB from epoch to epoch are normal because the validation set has only 300 mixtures.

7.2 Test-set metrics

Table 4 gives SI-SDR, STOI and wideband PESQ on the test set with 95% bootstrap confidence intervals. The headline row group excludes the 65 clean-scenario items, where the mixture already equals the reference.

Table 4: Test metrics from the earlier checkpoint, mean with 95% bootstrap CI. The headline column excludes the 65 clean-scenario items.

Metric	Non-clean (n=1135)	All items (n=1200)
SI-SDR input (dB)	0.28 [0.00, 0.56]	5.95 [4.62, 7.33]
SI-SDR output (dB)	11.06 [10.56, 11.56]	12.72 [12.09, 13.34]
SI-SDRi (dB)	**10.78 [10.34, 11.21]**	6.77 [5.73, 7.76]
STOI input	0.772 [0.765, 0.780]	0.785 [0.777, 0.792]
STOI output	0.872 [0.863, 0.880]	0.879 [0.870, 0.887]
PESQ output (wideband)	2.20 [2.15, 2.24]	2.33 [2.28, 2.38]

On non-clean items the model improves SI-SDR by 10.8 dB and STOI by 0.10. The all-items SI-SDRi of 6.77 dB is lower purely because of the clean scenario (Table 5): there the input SI-SDR is about 105 dB, so even a 41.6 dB output SI-SDR registers as a −63 dB "improvement". That is a metric artefact rather than a quality problem. STOI is 1.00 and PESQ is 4.62 on clean items, so clean speech passes through almost unchanged.

Table 5: Results by scenario from the earlier checkpoint. STOI is reported as input/output.

Scenario	n	SI-SDR in	SI-SDR out	SI-SDRi [95% CI]	STOI in/out	PESQ
noise	290	4.22	17.73	13.51 [12.79, 14.23]	0.90/0.95	2.89
speech	294	1.15	10.81	9.65 [8.70, 10.56]	0.76/0.88	2.29
speech_noise	551	−2.25	7.68	9.93 [9.30, 10.52]	0.71/0.83	1.78
clean	65	104.86	41.64	−63.22 [−64.93, −61.67]	1.00/1.00	4.62

Denoising alone (noise scenario) is easiest at 13.5 dB. Removing a competing talker (speech, 9.7 dB) and the hardest combination of talker plus noise (speech_noise, 9.9 dB) both give around 10 dB; the latter has the lowest output STOI (0.83) and PESQ (1.78). The SI-SDRi gain also shrinks as input SNR rises, from about 13 dB at low SNR to 9 dB at high SNR, which is expected because there is less noise left to remove; absolute quality (PESQ) improves with SNR. The model does not collapse at hard ratios, and it still gains about 12 dB at SIR below −3.3 dB and SNR below −1.7 dB. Across the 16 DEMAND environments the mean SI-SDRi ranges from 6.9 dB (restaurant, `PRESTO`) to 17.8 dB (washing, `DWASHING`); the hardest environments are crowded or transit scenes. The full SIR/SNR and environment breakdowns are in Section B.

7.3 Qualitative examples

Six test mixtures were drawn at random with a fixed seed and listened to. Table 6 lists SI-SDR before and after processing with the final checkpoint; all six improved by 11.3 to 21.4 dB. Figure 3 shows the spectrograms for the best of the six. These examples are illustrative and do not replace the statistics above.

Table 6: Randomly chosen test examples from the final checkpoint. The last column gives the change relative to the earlier checkpoint.

Test index	Input SI-SDR (dB)	Output SI-SDR (dB)	Improvement (dB)	vs. earlier (dB)
323	−3.50	12.27	+15.77	+1.35
761	3.80	21.57	+17.77	+0.25
611	2.52	17.88	+15.36	+1.00
369	−0.77	20.67	+21.44	+0.52
49	1.60	12.88	+11.27	+0.51
1016	−0.66	11.38	+12.04	+0.13

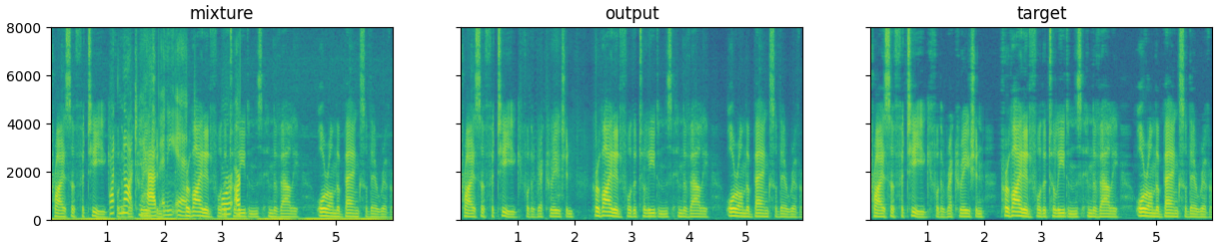


Figure 3: Test example 369 (+21.4 dB, the best of the six): mixture (left), model output (centre) and clean target (right). The competing voice and broadband noise are removed, and the output closely resembles the target.

7.4 Speaker confusion

The most characteristic TSE failure is extracting the *wrong* speaker. For the 845 test mixtures that contain an interferer we embedded the output, the clean target and a 5 s segment of the interferer with ECAPA-TDNN, and counted an output as confused when it is closer in cosine similarity to the interferer than to the target. The confusion rate is **7.7%** (65 of 845). Mean similarity to the target rises from 0.584 for the mixture to 0.772 for the output. Figure 4 (top) shows the two clusters and (bottom) the rate by condition: it is roughly independent of SIR (6.5–8.3%), slightly higher with added noise (about 8.4% versus 6.5%), and clearly depends on enrollment length (about 2% at 12 s versus 8–13% at 2–3 s; values read from the figure). Longer, cleaner enrollment is therefore the most effective lever against confusion.

7.5 Downstream intelligibility

To measure usefulness beyond signal metrics, 300 random test mixtures were transcribed with Whisper `small.en` [23], using Whisper's transcript of the *clean target* as the reference rather than human transcripts, and WER was computed with `jiwer`. Table 7 shows that extraction more than halves the WER.

The largest gain is with a competing talker: WER drops from 40.4% to 10.5% for speech and from 60.0% to 31.9% for speech plus noise. Noise-only mixtures are already easy for Whisper (7.0%), and clean speech is not damaged (0% WER).

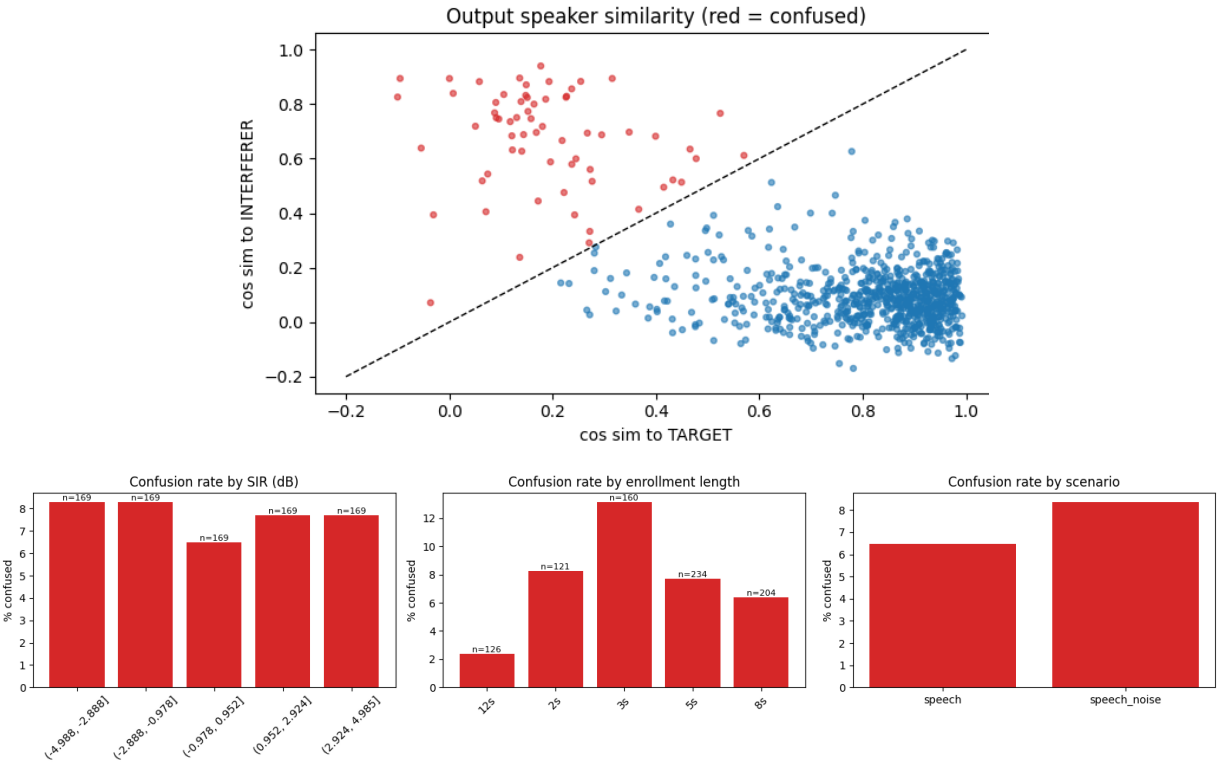


Figure 4: Top: output similarity to the target versus the interferer (red points are confused outputs). Bottom: confusion rate by SIR bin, enrollment length and scenario.

Table 7: Whisper WER (%) on 300 test mixtures from the final checkpoint.

Subset	n	Mixture	Extracted
All	300	38.3	**18.1**
Excluding clean	285	40.2	**18.9**
noise	77	7.0	5.8
speech	77	40.4	10.5
speech_noise	131	60.0	31.9
clean	15	0.0	0.0

Additional diagnostics are collected here. The model improves SI-SDR on 88.3% of the 1200 test mixtures. For the 845 mixtures with an interferer, the final checkpoint gives a mean SI-SDRi of 11.19 dB when the output matches the target speaker and −6.40 dB when it does not. The relative WER reduction is 52.9%. SI-SDRi correlates strongly negatively with input SI-SDR ($r = -0.90$), weakly negatively with SNR ($r = -0.21$) and SIR ($r = -0.11$), and essentially not with enrollment duration ($r = 0.05$).

7.6 Efficiency

Table 8 reports inference cost on an NVIDIA L4 for a 6 s clip, processed offline in one pass.

The fp32 pass takes 164.9 ms for this 6 s clip; fp16 saves memory but is slower at batch size one. These are offline GPU measurements. They do not imply low latency: the bidirectional LSTMs need the whole clip, and CPU speed has not yet been measured.

Table 8: Model size and inference cost for a 6 s clip, batch 1, on an NVIDIA L4.

	fp32	fp16 autocast
Parameters	16.83 M	
Compute	23.87 GMACs per clip (3.98 GMACs per second of audio)	
Time per clip	164.9 ms	181.1 ms
Offline RTF	0.0275	0.0302
Peak GPU memory	1057 MiB	870 MiB

7.7 Own-recording demonstration

The inference notebook also accepts a user's own files. A 13.4 s mixture with a 15.6 s enrollment recording was processed and the extracted audio was listened to. There is no reference signal, so no objective metric is given. This demo used the earlier checkpoint. **[TODO: rerun this demo with epoch 97 and add one sentence describing what we heard, especially whether the competing voice was reduced]**

8 Challenges and Limitations

- **Non-causal model.** Bidirectional LSTMs use future frames, so the current model cannot run with low latency. This departs from the proposal's low-latency goal, and a causal variant has not been built.
- **Size and compute.** At 16.83 M parameters and about 4 GMACs per second of audio, the model is not yet compatible with the compact CPU target. CPU speed is unmeasured, and the reported RTF is an offline GPU figure.
- **Speaker confusion.** In 7.7% of interferer mixtures the output matches the interferer better than the target, mostly with 2–3 s enrollments (Section 7.4). Target-absent behaviour is untested, so the system may extract the wrong voice when the target is silent.
- **No unseen-noise test.** DEMAND recordings are split by time, so the same 16 environments appear in training, validation and test. Performance on unseen noise types is therefore not demonstrated. Reverberation is also absent from both training and test.
- **Enrollment diversity.** Enrollments are VAD-cleaned from clean recordings, with only 10 recipes per training speaker. Real enrollments are noisier. The same energy VAD is used for the cached banks and for a user's own recording, as described in Section 4.
- **Length mismatch.** Training uses 4 s clips and evaluation 6 s clips. Long inputs, such as the 13.4 s demo, are processed in one pass or in 30 s chunks without overlap-add, which may cause boundary artefacts.
- **Evaluation status.** The 300-mixture validation set is small. The test SI-SDR/STOI/PESQ tables were computed with an earlier checkpoint and must be rerun for the final model. The clean scenario distorts the aggregate SI-SDRi, so non-clean results are the headline.
- **WER reference.** The WER check uses Whisper transcripts of the clean target as the reference, not human transcripts, and covers 300 mixtures.
- **Reproducibility.** The configured random seed is not applied inside the training loop, and dataset workers use unseeded generators, so reruns see different sampled mixtures.

9 Remaining Work and Revised Plan

Table 9 lists the remaining tasks with a primary owner and a priority. Three items are high priority: rerunning the test metrics with the epoch-97 checkpoint, the conditioning ablation promised in the proposal, and a test on unseen noise environments. CPU efficiency and a causal variant are medium priority and depend on the time left after the high-priority items.

Table 9: Remaining work and primary owner. Ownership follows the proposal's module assignments.

Task	Description	Owner	Priority
Final-checkpoint evaluation	Rerun SI-SDRi, STOI and PESQ (plus PESQ of the unprocessed mixture) with the epoch-97 checkpoint and refresh all breakdowns	Sostika	High
Conditioning ablation	Compare additive-only, FiLM-only and concatenation/gating, as promised in the proposal	Prashamsa	High
Unseen-noise test	Hold out whole DEMAND environments, or add MUSAN or self-recorded noise, for testing	Austina	High
Efficiency measurement	RTF, latency and memory on CPU (GPU RTF 0.028 already measured)	Sostika	Medium
Model compression and causal variant	Smaller configuration; unidirectional LSTM and causal STFT if time permits	Prashamsa, Sostika	Medium
Generalisation additions	Reverberation (RIRs), noisy and longer enrollments to cut speaker confusion, target-absent samples	Austina	Medium
Demo and documentation	Packaged inference script or small demo, final report and presentation	All	High

10 Conclusion

We moved from the proposal's causal Conv-TasNet design to a speaker-conditioned BSRNN, and training is complete. The model reached a validation SI-SDRi of 11.84 dB on unseen speakers. In the earlier-checkpoint test evaluation, non-clean SI-SDR improved by about 10.8 dB and STOI rose from 0.77 to 0.87; the final-checkpoint WER check fell from 38.3% to 18.1%. The trade-off is clear: the current model is larger and non-causal, and 7.7% of interferer mixtures were classified as speaker confusions. We still need the final test-metric rerun, the conditioning ablation, an unseen-noise test and CPU measurements. On the team side, Austina led data preparation, Prashamsa led the speaker encoder and conditioning work, and Sostika led training and inference; all three shared the literature review, evaluation and writing.

References

[1] Q. Wang, H. Muckenhirn, K. Wilson, P. Sridhar, Z. Wu, J. Hershey, R. A. Saurous, R. J. Weiss, Y. Jia, and I. L. Moreno, "VoiceFilter: Targeted voice separation by speaker-conditioned spectrogram masking," in *Proc. Interspeech*, 2019.

[2] H. Sato, T. Moriya, M. Mimura, S. Horiguchi, T. Ochiai, T. Ashihara, A. Ando, K. Shinayama, and M. Delcroix, "SpeakerBeam-SS: Real-time target speaker extraction with lightweight Conv-TasNet and state space modeling," in *Proc. Interspeech*, 2024. arXiv:2407.01857.

[3] Y. Luo and J. Yu, "Music source separation with band-split RNN," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 1893–1901, 2023.

[4] Y. Luo, Z. Chen, and T. Yoshioka, "Dual-path RNN: Efficient long sequence modeling for time-domain single-channel speech separation," in *Proc. IEEE ICASSP*, 2020.

[5] J. Yu and Y. Luo, "Efficient monaural speech enhancement with universal sample rate band-split RNN," in *Proc. IEEE ICASSP*, 2023.

[6] Z.-Q. Wang, S. Cornell, S. Choi, Y. Lee, B.-Y. Kim, and S. Watanabe, "TF-GridNet: Integrating full- and sub-band modeling for speech separation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 3221–3236, 2023.

[7] B. Desplanques, J. Thienpondt, and K. Demuynck, "ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification," in *Proc. Interspeech*, pp. 3830–3834, 2020.

[8] M. Ravanelli *et al.*, "SpeechBrain: A general-purpose speech toolkit," 2021. arXiv:2106.04624.

[9] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville, "FiLM: Visual reasoning with a general conditioning layer," in *Proc. AAAI*, 2018.

[10] J. Le Roux, S. Wisdom, H. Erdogan, and J. R. Hershey, "SDR – half-baked or well done?," in *Proc. IEEE ICASSP*, pp. 626–630, 2019.

[11] R. Yamamoto, E. Song, and J.-M. Kim, "Parallel WaveGAN: A fast waveform generation model based on generative adversarial networks with multi-resolution spectrogram," in *Proc. IEEE ICASSP*, pp. 6199–6203, 2020.

[12] C. H. Taal, R. C. Hendriks, R. Heusdens, and J. Jensen, "An algorithm for intelligibility prediction of time–frequency weighted noisy speech," *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 19, no. 7, pp. 2125–2136, 2011.

[13] ITU-T, "Recommendation P.862.2: Wideband extension to recommendation P.862 for the assessment of wideband telephone networks and speech codecs," tech. rep., International Telecommunication Union, 2007.

[14] C. K. A. Reddy *et al.*, "The INTERSPEECH 2020 deep noise suppression challenge: Datasets, subjective testing framework, and challenge results," in *Proc. Interspeech*, 2020.

[15] H. Schröter, A. N. Escalante-B., T. Rosenkranz, and A. Maier, "DeepFilterNet: A low complexity speech enhancement framework for full-band audio based on deep filtering," in *Proc. IEEE ICASSP*, 2022.

[16] Y. N. Dauphin, A. Fan, M. Auli, and D. Grangier, "Language modeling with gated convolutional networks," in *Proc. ICML*, 2017.

[17] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "LibriSpeech: An ASR corpus based on public domain audio books," in *Proc. IEEE ICASSP*, pp. 5206–5210, 2015.

[18] J. Thiemann, N. Ito, and E. Vincent, "The diverse environments multi-channel acoustic noise database (DEMAND)," *Proceedings of Meetings on Acoustics*, vol. 19, p. 035081, 2013.

[19] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2019.

[20] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," in *Proc. ICLR*, 2017.

[21] P. Micikevicius *et al.*, "Mixed precision training," in *Proc. ICLR*, 2018.

[22] T. Chen, B. Xu, C. Zhang, and C. Guestrin, "Training deep nets with sublinear memory cost," 2016. arXiv:1604.06174.

[23] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. ICML*, 2023.

A Reproducibility Notes

- **Code:** `tselib.py` contains the model, sampler, datasets, loss and training loop, with two YAML configuration files. These are kept on Google Drive under `tse-project/`. **[TODO: add the team's GitHub repository URL, if one exists]**
- **Training notebook:** downloads LibriSpeech and DEMAND, loads manifests and cached embedding banks, patches gradient checkpointing, and calls `T.train(...)`.
- **Test notebook:** loads the best checkpoint, plays the mixture, output and target, computes SI-SDR, STOI and PESQ on the test manifest, and separates user-supplied files.
- **Software:** PyTorch 2.x (CUDA), SpeechBrain, `pystoi`, `pesq`, `jiwer` and Whisper through `transformers` for the WER check.
- **Analysis cells:** per-scenario, SNR, SIR and noise-environment breakdowns, bootstrap confidence intervals, speaker-confusion analysis, WER and FLOP/RTF benchmarking are in the training notebook after the test-loading cell.

B Additional Test Breakdowns

Table 10: SI-SDRi (dB) by difficulty from the earlier checkpoint. SIR bins use the 845 mixtures with an interferer; SNR bins use the 841 mixtures with noise.

SIR bin (dB)	n	SI-SDRi	PESQ	SNR bin (dB)	n	SI-SDRi	PESQ
$[-5.0, -3.3]$	131	11.98	1.81	$[-5.0, -1.7]$	155	12.86	1.77
$(-3.3, -1.7]$	152	9.03	1.80	$(-1.7, 1.7]$	136	13.05	2.05
$(-1.7, 0.0]$	134	10.57	1.91	$(1.7, 5.0]$	142	11.75	2.22
$(0.0, 1.7]$	147	10.22	1.99	$(5.0, 8.3]$	135	10.83	2.31
$(1.7, 3.3]$	144	8.42	2.05	$(8.3, 11.6]$	126	9.05	2.27
$(3.3, 5.0]$	137	9.03	2.19	$(11.6, 14.9]$	147	9.20	2.42

Table 11: Mean SI-SDRi and output quality per DEMAND environment from the earlier checkpoint (mixtures containing noise only).

Env.	n	SI-SDRi	STOI	PESQ	Env.	n	SI-SDRi	STOI	PESQ
DKITCHEN	57	13.87	0.92	2.56	OOFFICE	48	13.22	0.89	2.35
DLIVING	61	9.14	0.85	1.79	PRESTO	49	6.87	0.79	1.64
DWASHING	33	17.76	0.92	2.77	PSTATION	64	8.56	0.84	1.97
NFIELD	44	12.86	0.88	2.26	SPSQUARE	49	9.45	0.85	1.93
NPARK	52	11.14	0.85	1.90	STRAFFIC	51	10.48	0.89	2.27
NRIVER	44	12.01	0.86	2.08	TBUS	63	13.38	0.92	2.62
OHALLWAY	53	13.40	0.88	2.33	TCAR	52	13.52	0.92	2.82
OMEETING	65	8.38	0.83	1.82	TMETRO	56	8.34	0.83	1.81